

ZAGREB MAKSIMIR, Croatia

WMO: 142400

Lat: **45.8219N**Lon: **16.0336E**Elev: **128**StdP: **99.80**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-6.7	-13.0	1.2	-8.2	-10.3	1.6	-4.4	7.1	-0.6	6.1	0.2	1.6	30	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.9	21.7	31.2	21.2	29.6	20.5	22.7	30.2	21.9	29.3	21.2	28.2	2.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.2	26.1	19.4	14.4	25.3	18.6	13.7	24.6	67.5	30.1	64.4	29.4	61.7	28.3	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.6	4.7	4.1	-12.6	35.0	3.4	2.2	-15.1	36.5	-17.1	37.8	-19.0	39.0	-21.5	40.6		
(5)			-12.9	24.3	3.3	1.0	-15.3	25.1	-17.3	25.7	-19.2	26.3	-21.6	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	1.3	3.3	7.5	12.3	16.8	20.6	22.2	22.0	17.0	12.3	7.2	2.1
	DBStd	8.36	4.56	4.57	4.34	3.77	3.45	3.44	3.03	3.06	3.36	4.01	4.41	4.35
	HDD10.0	958	271	189	100	21	1	0	0	0	0	24	105	247
	HDD18.3	2657	528	420	336	182	71	17	4	6	64	191	335	503
	CDD10.0	1722	1	3	23	91	212	317	378	371	211	94	20	2
	CDD18.3	379	0	0	0	1	24	84	124	118	25	2	0	0
	CDH23.3	3632	0	0	1	28	237	814	1226	1107	201	18	0	0
	CDH26.7	1256	0	0	0	3	44	268	470	431	41	0	0	0
Wind	WSAvg	1.6	1.5	1.7	2.0	1.9	1.8	1.7	1.6	1.5	1.5	1.4	1.4	1.4
Precipitation	PrecAvg	1001	103	56	88	67	80	87	73	91	88	93	96	99
	PrecMax	1769	711	174	400	141	219	184	186	263	214	304	235	403
	PrecMin	558	3	4	9	15	17	18	26	14	30	4	9	11
	PrecStd	326	145	41	85	30	50	38	34	51	43	63	59	99
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	17.5	21.9	26.5	29.9	33.4	35.1	36.0	30.3	25.3	20.4	15.5
		MCWB	10.1	10.1	12.5	16.0	19.3	22.0	21.9	21.2	19.9	17.8	14.2	11.2
	2%	DB	12.6	14.7	19.3	23.6	27.8	31.4	32.8	32.8	27.6	23.0	17.5	12.7
		MCWB	8.9	9.1	11.5	14.4	18.3	21.5	22.0	21.5	19.4	16.9	13.1	9.6
	5%	DB	10.5	12.7	17.1	21.7	26.0	29.8	31.0	31.1	25.7	20.8	15.5	10.7
		MCWB	7.7	8.2	10.5	13.6	17.4	20.7	21.3	20.9	18.4	15.8	12.1	8.3
	10%	DB	8.2	10.6	14.9	19.7	24.1	28.0	29.3	29.0	23.6	18.6	13.6	8.3
		MCWB	6.1	6.9	9.5	12.6	16.5	19.8	20.6	20.4	17.4	14.8	11.3	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.9	13.1	16.8	20.6	23.6	24.0	23.7	21.1	18.6	15.0	11.6
		MCDB	14.0	16.3	20.1	24.9	28.0	30.9	31.8	31.5	27.8	23.9	18.4	14.6
	2%	WB	9.2	9.6	12.0	15.2	19.2	22.3	22.7	22.5	19.9	17.4	13.7	10.0
		MCDB	12.1	14.0	18.1	22.0	26.2	29.6	30.4	30.1	26.1	21.8	16.6	12.6
	5%	WB	7.8	8.5	11.0	14.1	18.1	21.3	21.9	21.6	19.0	16.3	12.5	8.4
		MCDB	10.2	12.1	15.9	19.8	24.6	28.3	29.4	29.0	24.4	19.9	15.2	10.4
	10%	WB	6.2	7.2	10.0	13.2	17.1	20.4	21.1	20.8	18.1	15.2	11.3	6.8
		MCDB	8.1	10.2	14.5	18.6	23.0	27.0	28.1	27.8	22.5	18.2	13.6	8.3
Mean Daily Temperature Range	5% DB	MDBR	6.4	8.1	9.6	10.7	10.7	10.8	11.1	11.2	9.8	9.0	6.9	5.9
		MCDBR	10.4	12.6	13.8	14.7	13.8	13.3	13.9	14.7	13.0	12.1	10.7	9.8
		MCWBR	6.9	7.6	7.3	6.9	5.9	5.3	5.1	5.3	5.7	6.5	6.5	6.8
	5% WB	MCDBR	9.6	11.3	12.2	12.7	12.6	12.4	12.7	13.0	11.7	11.0	9.6	9.1
		MCWBR	6.5	7.1	6.6	6.1	5.7	5.3	5.1	5.3	5.4	6.1	6.1	6.6
Clear-Sky Solar Irradiance	taub	0.335	0.361	0.398	0.447	0.435	0.439	0.439	0.440	0.425	0.397	0.363	0.335	
	taud	2.348	2.287	2.229	2.112	2.195	2.228	2.227	2.237	2.246	2.305	2.350	2.385	
	Ebn at Noon	751	801	822	811	835	832	826	809	784	749	709	710	
	Edh at Noon	77	99	121	148	141	137	135	129	118	97	77	68	
All-Sky Solar Radiation	RadAvg	1.23	1.99	3.19	4.37	5.40	6.07	6.10	5.37	3.71	2.43	1.34	1.00	
	RadStd	0.19	0.33	0.43	0.50	0.50	0.44	0.40	0.58	0.48	0.28	0.18	0.16	